

NYC Airbnb

Machine Learning Analysis

Regression • Classification • Clustering

Our Team

1

Regression Model

Random Forest vs XGBoost

2

Classification Model

Room Type Prediction

3

Clustering Model

Price-based Groups

Today's Agenda

1 **Introduction**

Data Preprocessing

2 **Regression**

Price Prediction

3 **Classification**

Room Type Prediction

4 **Clustering**

Price Groups

Dataset Overview



20,560

Total Listings



22 → 13

Features



**NYC 5
Boroughs**

Location



2024

Year

“Data preprocessing is critical for model success”

Data Preprocessing

Problem: Multicollinearity

Pair	Correlation
reviews_per_month ↔ reviews_ltm	0.85
bedrooms ↔ beds	0.70

Solution: Remove Redundant

Remove
✗ reviews_ltm
✗ beds



Final 13 Features:

Location (3) • Facilities (3) • Reviews (4) • Others (3)

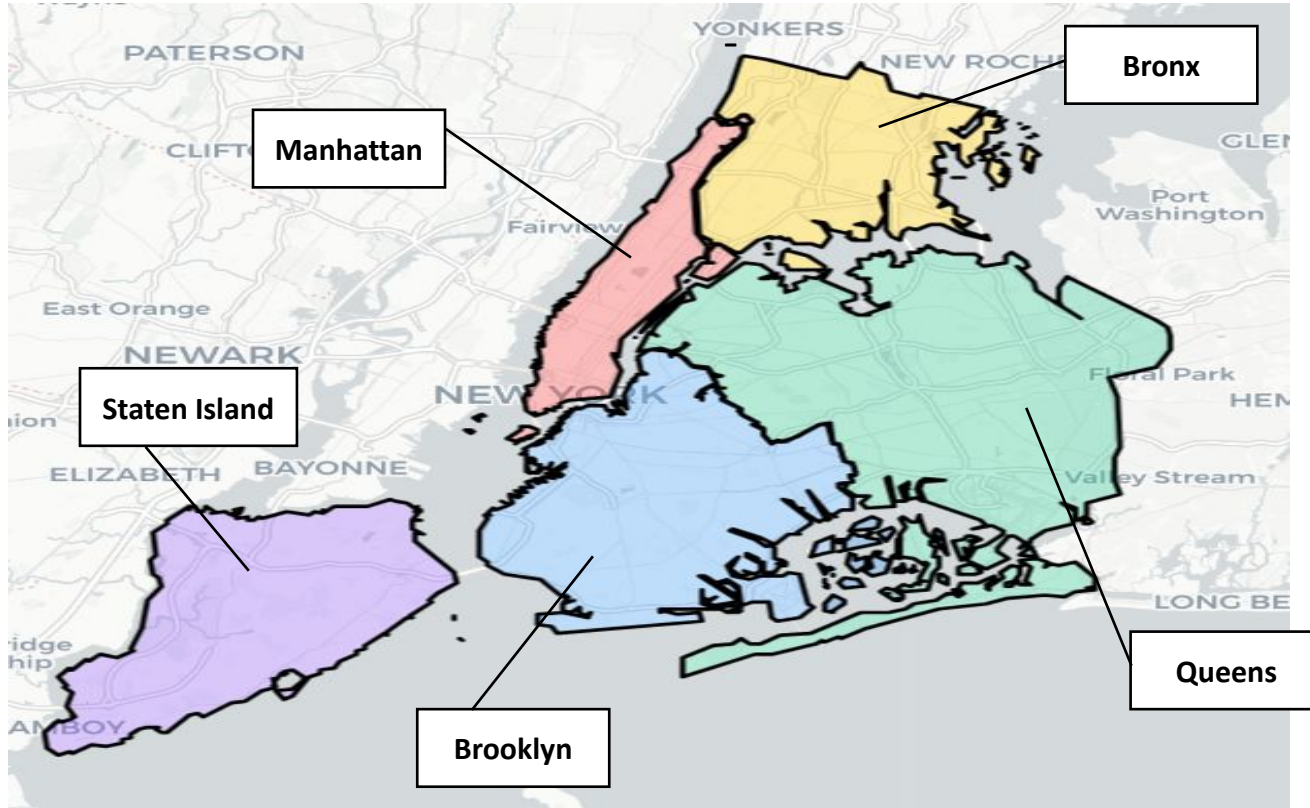
Clean, preprocessed data ready for ML models

PART 1

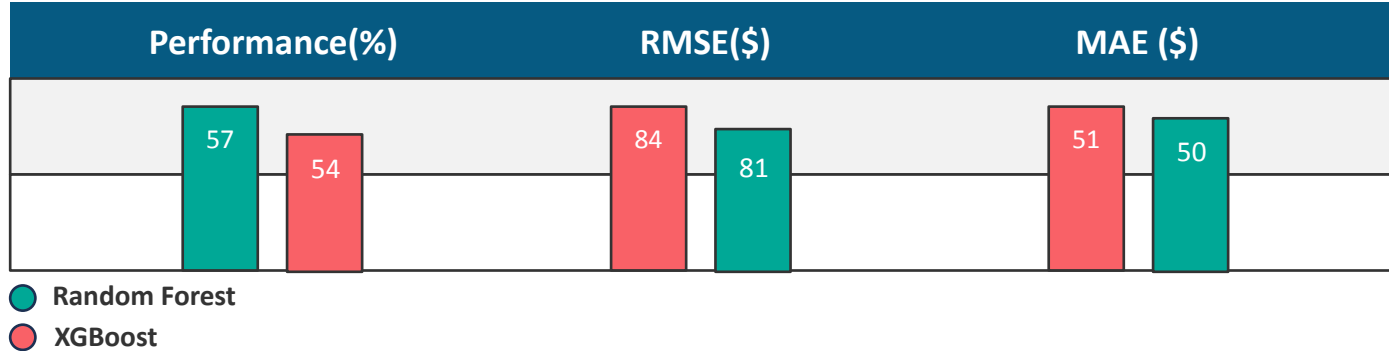
Regression Model

Price Prediction

Data Overview



Regression: Random Forest vs XGBoost



Analysis

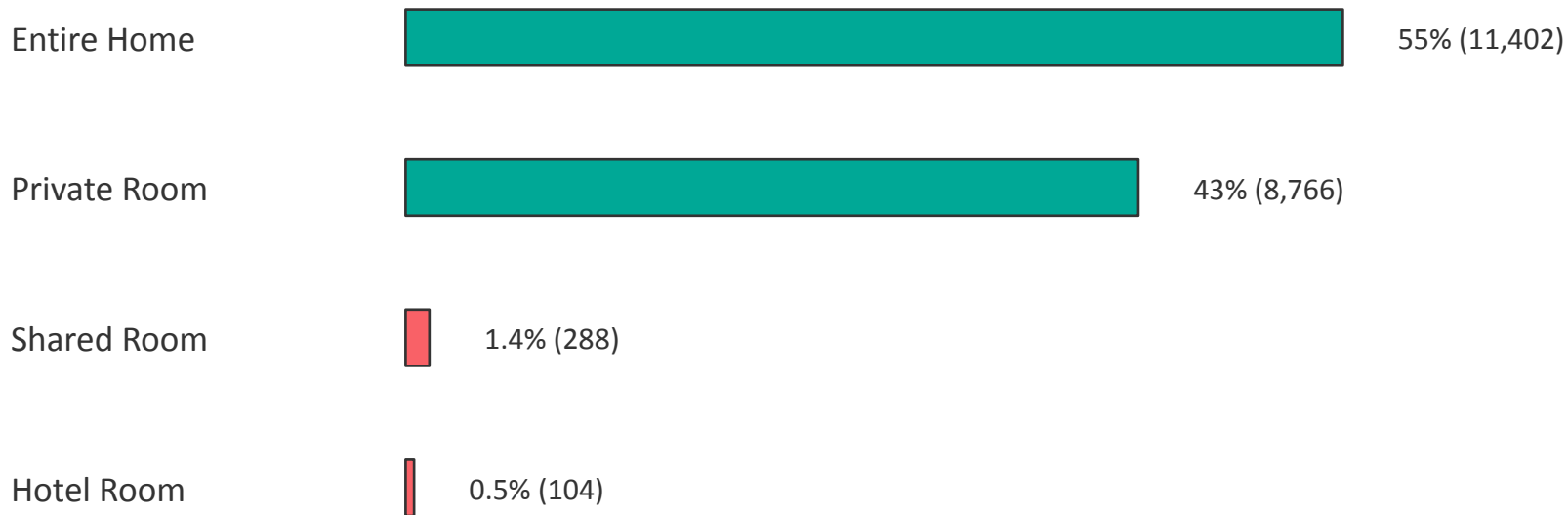
*"XGBoost has a **lower MAE(50), RMSE (81)**. This means its predictions are closer to the actual values than Random Forest."*

PART 2

Classification Model

Room Type Prediction

The Challenge: Class Imbalance



Problem: 109:1 ratio! (Most vs Least)

Solution: Class Weights

Give more penalty when model gets minority class wrong

1.0x Entire Home	1.3x Private Room	39.6x Shared Room	109.6x Hotel Room
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Hotel Room mistake = 110x worse than Entire Home!

“Class imbalance needs special handling (weights)”

Classification Results

Accuracy

Random Forest

75.7%

XGBoost

76.7%

F1-Score

Random Forest

0.555

XGBoost

0.565

"XGBoost outperformed Random Forest in both Accuracy and F1-Score metrics."

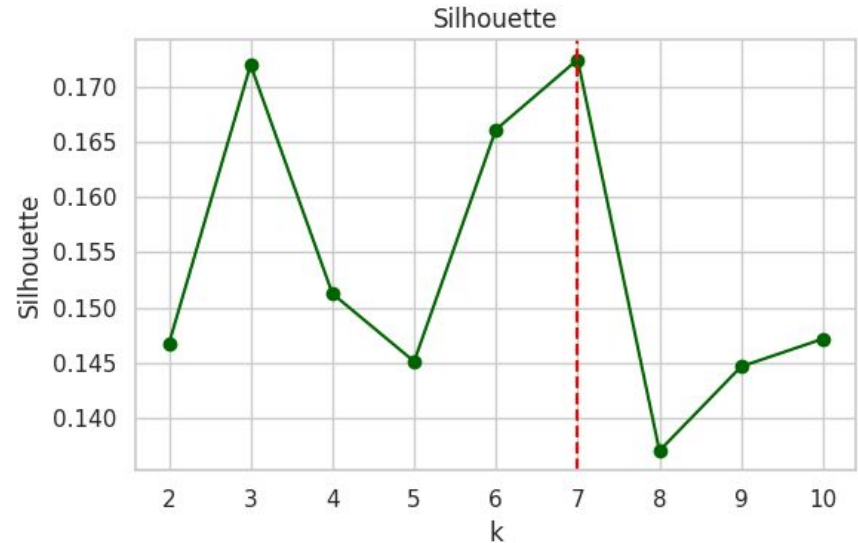
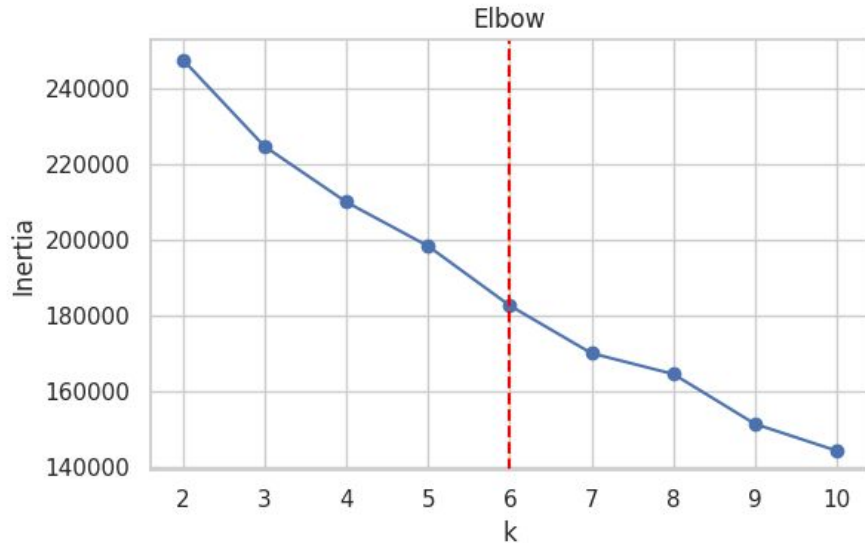
PART 3

Clustering Model

Price-based Grouping

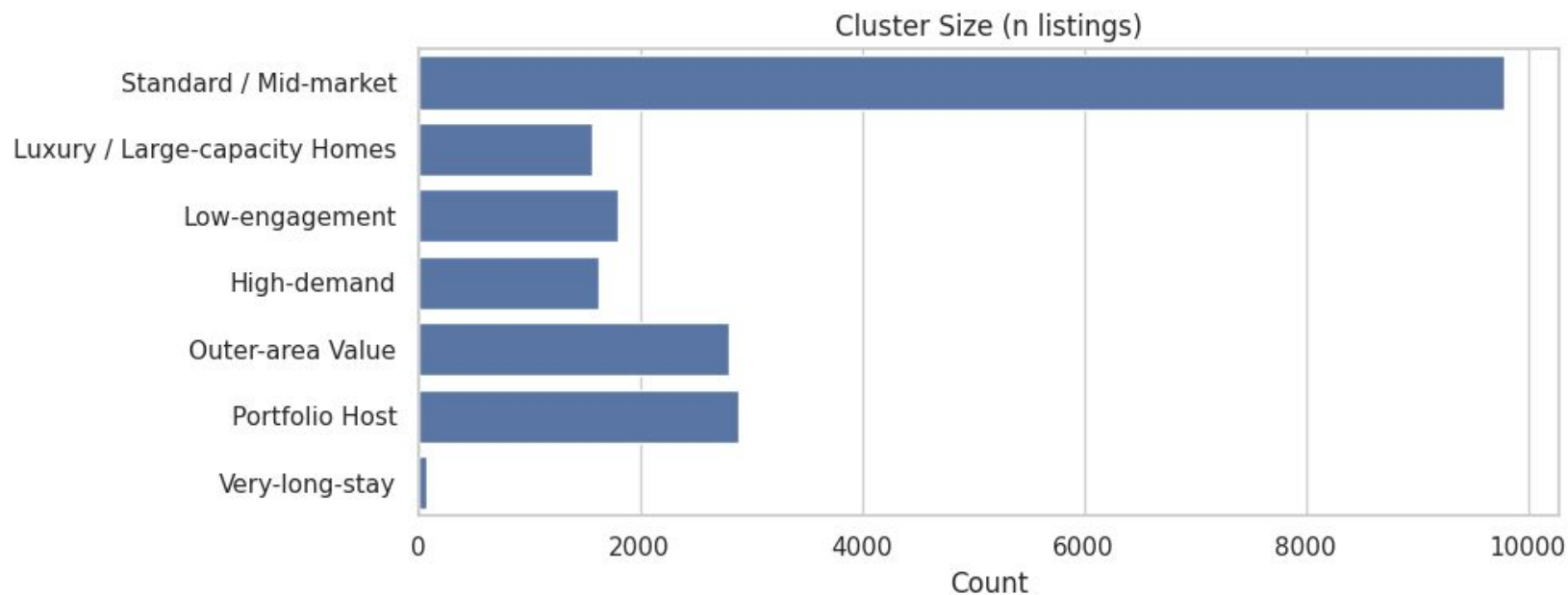
Unsupervised Learning: K-Means

Goal: Determining the optimal number of clusters



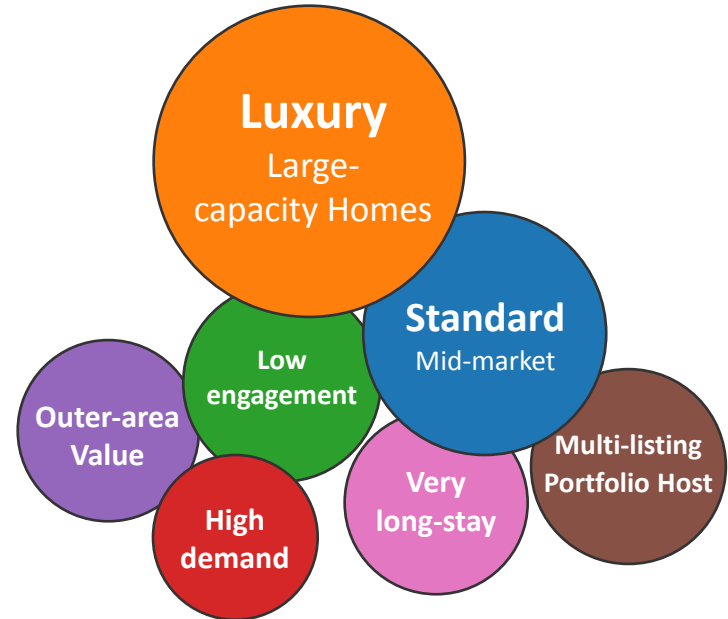
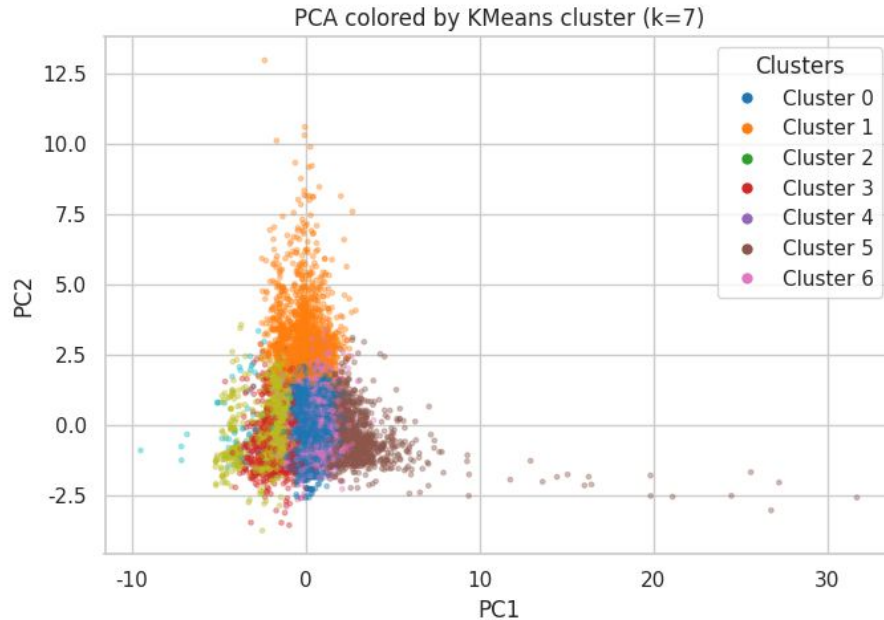
Unsupervised Learning: K-Means

Goal: Group similar-priced listings together



Unsupervised Learning: K-Means

“Clustering reveals hidden patterns in data”



Thank You!

Questions?